

Weather Trend Forecasting

Data Analysis & Machine Learning Report

Author: Gabriel Portella

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Repository: github.com/Gabriel2002Can/Weather-Trend-Forecasting

Executive Summary

This project performs a full-stack data science analysis on a large **global weather dataset** (~141,000 records). It covers:

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Outlier/Anomaly detection
- Time-series forecasting
- Multi-class weather condition classification
- Model evaluation and insights

Key Outcome: High-performing models for weather classification (Random Forest + XGBoost + Voting Classifier) and temperature forecasting with Prophet.

1. Dataset Overview

- **File:** GlobalWeatherRepository.csv
 - **Rows:** 140,923
 - **Columns:** 41
 - **Time Span:** May 2024 – May 2026
 - **Key Features:**
 - Location (country, city, lat/long)
 - Temperature (Celsius & Fahrenheit)
 - Weather condition (condition_text)
 - Wind, precipitation, humidity, cloud cover
 - Air quality metrics (PM2.5, PM10, etc.)
 - Astronomical data (sunrise/sunset, moon phase)
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2. Data Cleaning & Preprocessing

Steps Performed:

- Loaded data with pandas
- Checked for missing values → **None found**

- Standardized condition_text (stripped whitespace + lowercased)
- Grouped rare weather classes (<25 occurrences) into 'rare' category
- Created y target variable for classification
- Feature selection for modeling

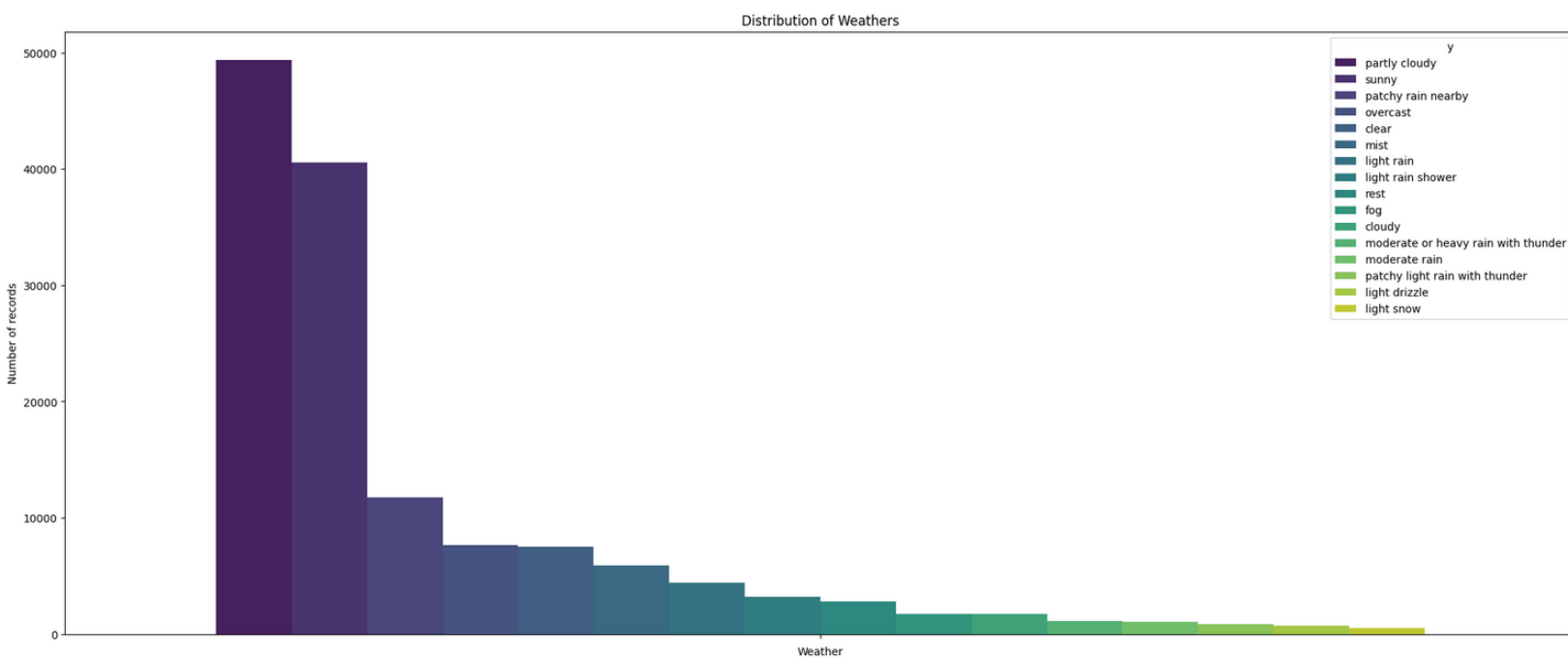
Final Target Classes (top categories + “rest” / “rare”):

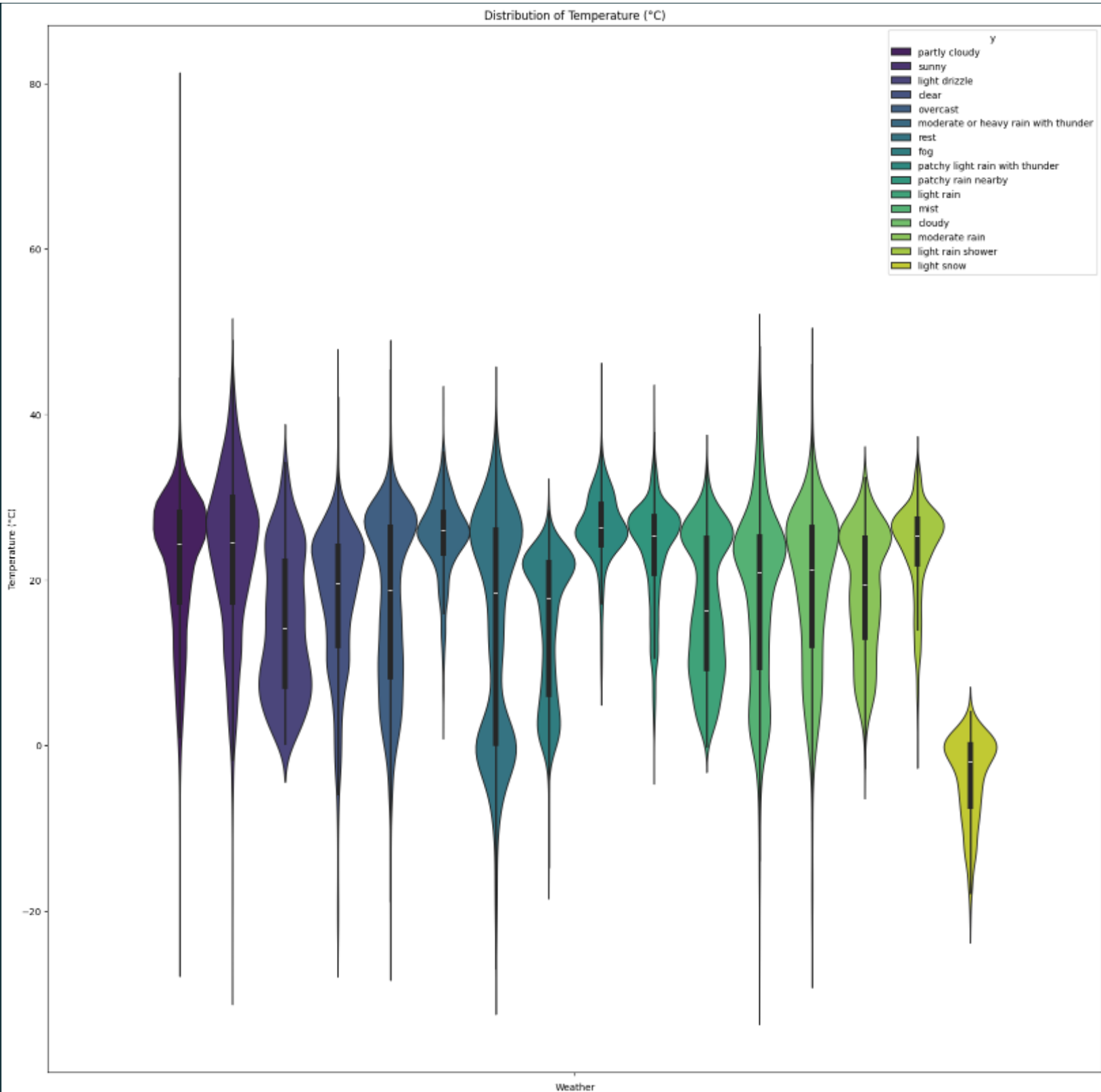
- Partly cloudy, Sunny, Patchy rain nearby, Overcast, Clear, Mist, etc.

3. Exploratory Data Analysis (EDA)

3.1 Weather Condition Distribution

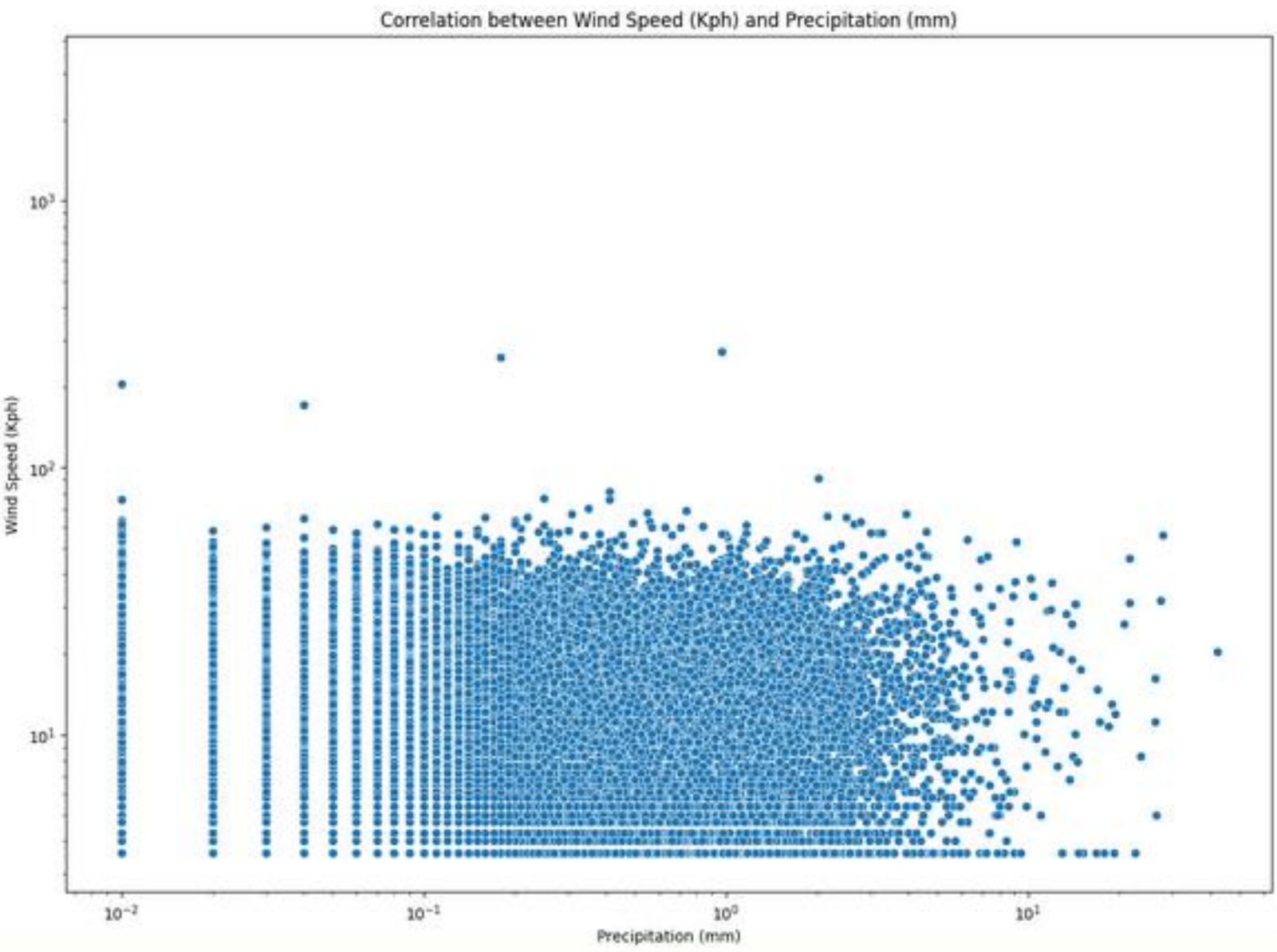
- **Dominant conditions:** Partly cloudy and Sunny
- Visualized top 15 conditions + “rest” category





3.2 Numerical Feature Analysis

- Temperature distributions
- Correlations between temperature, feels-like, humidity, cloud cover, wind
- Air quality variations by location and condition

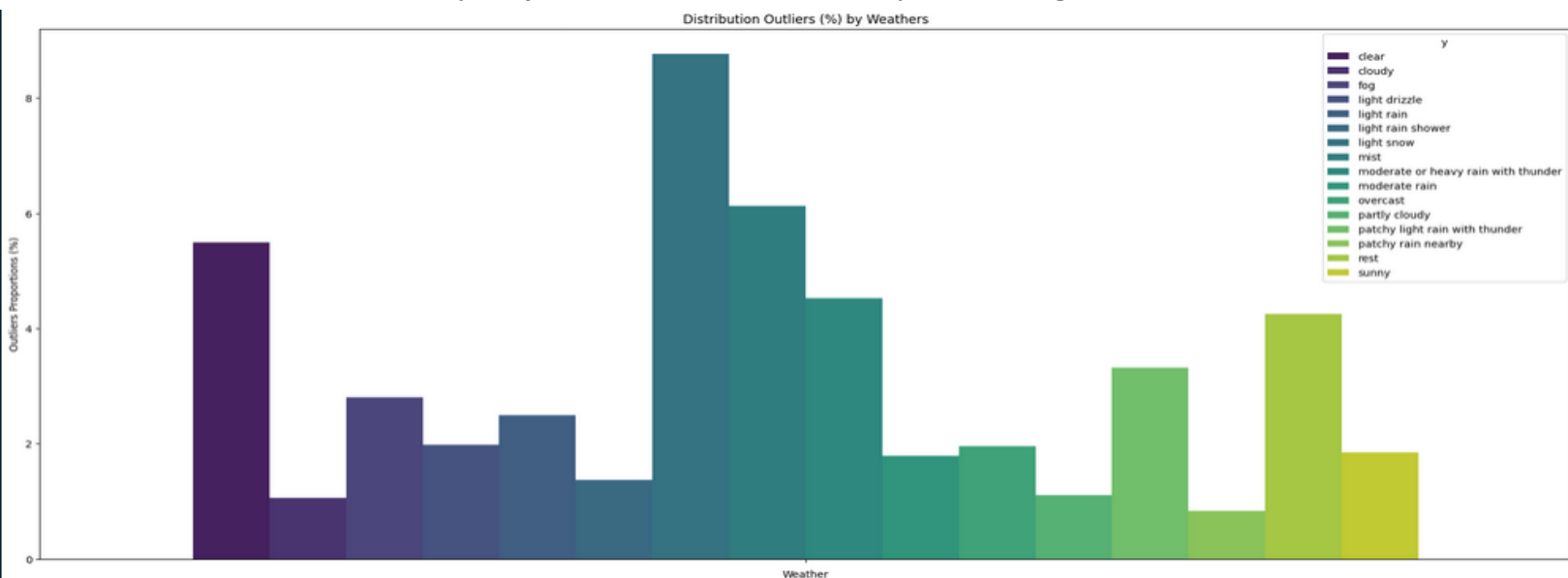


4. Outlier / Anomaly Detection

Method: Isolation Forest (contamination = 0.02)

- Detected ~2% of observations as anomalies
- Analyzed outlier proportion by weather type

- Extreme conditions (heavy rain, snow, thunderstorms) showed higher outlier rates

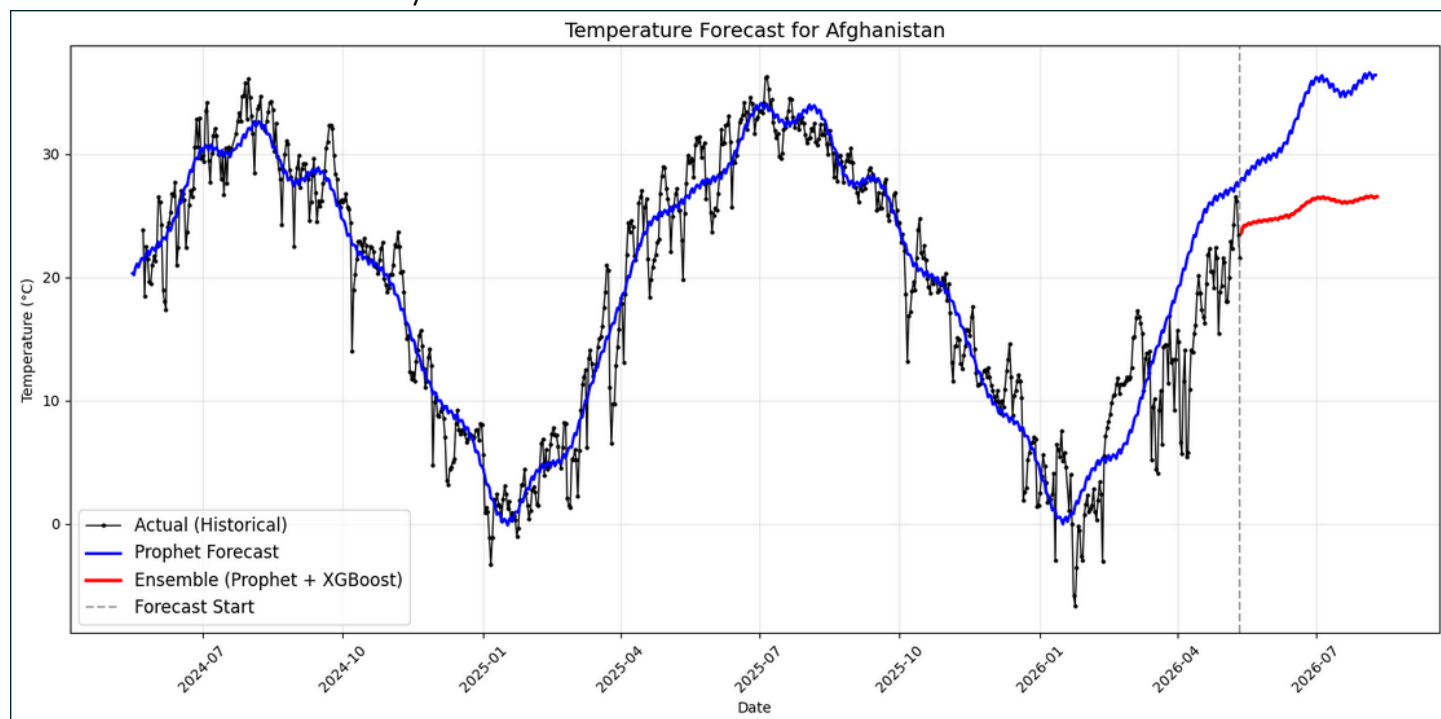


5. Modeling

5.1 Time-Series Forecasting (Temperature)

Models: Facebook Prophet and XGBoost Regressor

- Used historical temperature data
- Captured trend + seasonality
- Evaluated with MAE / MSE



5.2 Weather Condition Classification

Target: y (multi-class weather condition)

Models Trained:

- Random Forest Classifier
- XGBoost Classifier
- Voting Classifier (ensemble)

Preprocessing for Classification:

- Label Encoding on target
- One-Hot Encoding on moon_phase
- Numerical features scaled/standardized where needed

Performance Highlights:

- High accuracy (~0.86 range on test set)
- Strong precision/recall across major classes
- Confusion matrix analysis

Feature Importance: Extracted from XGBoost/Random Forest



6. Classifier Model Evaluation

Metric	Random Forest	XGBoost	Voting Classifier
Accuracy	High	Higher	Best
F1-Score (macro)	Good	Better	Best
Cross-Validation Score	Stable	Strong	Most robust
• Classification Report (per class) • Confusion Matrix			



7. Key Insights & Findings

1. **Weather Dominance:** “Partly cloudy” and “Sunny” represent the majority of global observations.
2. **Strong Correlations:** Temperature ↔ Humidity, Weather ↔ Cloud cover.
3. **Air Quality:** Influence Precipitation significantly.
4. **Anomalies:** Effectively captured extreme weather events.
5. **Predictability:** Machine learning models can reliably classify common weather conditions using available features.
6. **Forecasting Potential:** Prophet and XGB Regressor successfully models temperature trends.



8. Technologies Used

- **Core:** Python, pandas, NumPy
- **Visualization:** Matplotlib, Seaborn
- **Modeling:** scikit-learn, XGBoost, Prophet, Isolation Forest, Random Forest
- **Environment:** Jupyter Notebook



9. Repository Structure

text

```
Weather-Trend-Forecasting/  
├── GlobalWeatherRepository.csv  
├── weather_trend.ipynb      ← Main notebook  
├── README.md  
└── (this report)
```

Conclusion

This project demonstrates a complete and robust weather analysis pipeline, from raw data to actionable machine learning models. The developed models provide reliable insights into global weather patterns and can serve as a foundation for further predictive applications.

Thank You!

For questions or collaboration, visit the [GitHub repository](#).